

EXPERIMENT 3

EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION

Aim

To identify and handle missing values, duplicate records, outliers, inconsistent data, and perform data transformation techniques such as normalization and encoding using Python Pandas.

Software Requirements

- Python 3.x
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Scikit-Learn
 - Matplotlib
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Theory

Real-world datasets often contain:

- Missing Values
- Duplicate Records
- Incorrect Data Types
- Outliers
- Inconsistent Formatting

Data cleaning improves data quality and increases the accuracy of analytical models.

Data transformation converts data into suitable formats for analysis and machine learning.

Learning Outcomes

After completing this experiment students will be able to:

- ✓ Detect missing values
- ✓ Handle null values
- ✓ Remove duplicates
- ✓ Identify outliers

- ✓ Normalize data
 - ✓ Encode categorical variables
 - ✓ Prepare datasets for analytics
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Sample Dataset

Employee Dataset

EmpID	Name	Age	Salary	Department
-------	------	-----	--------	------------

101	Arun	25	25000	HR
102	Bala	NaN	30000	IT
103	Divya	28	NaN	HR
104	Priya	25	25000	HR
104	Priya	25	25000	HR
105	Ravi	60	150000	Finance

Algorithm 1

Identifying Missing Values

Step 1

Import pandas library

Step 2

Load dataset

Step 3

Use isnull() function

Step 4

Count missing values

Step 5

Display result

Program 1

```
import pandas as pd
```

```
df = pd.read_csv("employee.csv")
```

```
print(df.isnull())
```

Output

EmpID	Name	Age	Salary	Department
-------	------	-----	--------	------------

False	False	False	False	False
-------	-------	-------	-------	-------

False	False	True	False	False
-------	-------	------	-------	-------

False	False	False	True	False
-------	-------	-------	------	-------

False	False	False	False	False
-------	-------	-------	-------	-------

False	False	False	False	False
-------	-------	-------	-------	-------

False	False	False	False	False
-------	-------	-------	-------	-------

Program 2

Count Missing Values

```
print(df.isnull().sum())
```

Output

Column	Missing Values
--------	----------------

EmpID	0
-------	---

Name	0
------	---

Age	1
-----	---

Salary	1
--------	---

Department	0
------------	---

Algorithm 2

Replacing Missing Values

Step 1

Calculate mean value

Step 2

Replace null values

Step 3

Verify replacement

Program 3

```
df["Age"].fillna(df["Age"].mean(), inplace=True)

df["Salary"].fillna(df["Salary"].mean(), inplace=True)

print(df)
```

Output

EmpID	Name	Age	Salary	Department
-------	------	-----	--------	------------

101	Arun	25	25000	HR
102	Bala	32.6	30000	IT
103	Divya	28	51000	HR
104	Priya	25	25000	HR
104	Priya	25	25000	HR
105	Ravi	60	150000	Finance

Algorithm 3

Removing Duplicate Records

Step 1

Use duplicated()

Step 2

Count duplicate records

Step 3

Remove duplicates

Step 4

Verify dataset

Program 4

```
print(df.duplicated())

df.drop_duplicates(inplace=True)
```

```
print(df)
```

Output

Before Removal

Record Number Duplicate

1	False
2	False
3	False
4	False
5	True
6	False

After Removal

Total Records

5

Algorithm 4

Detecting Outliers

Step 1

Calculate Quartiles

Step 2

Compute IQR

Step 3

Calculate lower and upper limits

Step 4

Identify outliers

Formula

Inter Quartile Range

$$IQR = Q_3 - Q_1$$

Outlier Limits

Lower Limit = $Q_1 - 1.5 \times IQR$

Upper Limit = $Q_3 + 1.5 \times IQR$

Program 5

```
Q1 = df['Salary'].quantile(0.25)
```

```
Q3 = df['Salary'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5*IQR
```

```
upper = Q3 + 1.5*IQR
```

```
outliers = df[(df['Salary'] < lower) |  
              (df['Salary'] > upper)]
```

```
print(outliers)
```

Output

EmpID	Name	Salary
-------	------	--------

105	Ravi	150000
-----	------	--------

Visualization of Outlier

Employee Salary Analysis

Salary distribution showing potential Outlier.

Algorithm 5

Normalization

Purpose

Convert numerical values into a common scale.

Formula

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Program 6

```
from sklearn.preprocessing import MinMaxScaler
```

```
scaler = MinMaxScaler()
```

```
df[['Salary']] = scaler.fit_transform(df[['Salary']])  
  
print(df[['Salary']])
```

Output

Salary

0.00

0.04

0.21

0.00

1.00

Algorithm 6

Label Encoding

Purpose

Convert categorical data into numerical format.

Program 7

```
from sklearn.preprocessing import LabelEncoder  
  
encoder = LabelEncoder()  
  
df['Department'] = encoder.fit_transform(df['Department'])  
  
print(df)
```

Output

Department Encoded Value

Finance 0

HR 1

IT 2

Program 8

One-Hot Encoding

```
pd.get_dummies(df['Department'])
```

Output

Finance HR IT

0	1	0
0	0	1
0	1	0
0	1	0
1	0	0

Data Transformation Summary

Technique	Purpose
Fill Missing Values	Handle null data
Drop Duplicates	Remove redundancy
Outlier Detection	Improve data quality
Normalization	Scale data
Label Encoding	Convert categories
One-Hot Encoding	Binary representation

Applications

Healthcare

Cleaning patient records

Banking

Removing duplicate transactions

Education

Student performance analysis

Retail

Sales data preprocessing

Social Media

User behavior analysis

Advantages

1. Improves Data Quality
 2. Increases Model Accuracy
 3. Reduces Errors
 4. Removes Noise
 5. Enables Better Analytics
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Viva Questions

1. What is Data Cleaning?

The process of correcting inaccurate or incomplete data.

2. What are Missing Values?

Values that are absent from the dataset.

3. What is the purpose of fillna()?

To replace missing values.

4. What is a duplicate record?

A repeated row in a dataset.

5. Which function removes duplicates?

drop_duplicates()

6. What is an outlier?

An observation significantly different from others.

7. What is normalization?

Scaling numerical values to a standard range.

8. What is Min-Max Scaling?

A normalization technique that converts values between 0 and 1.

9. What is Label Encoding?

Converting categorical values into numeric labels.

10. What is One-Hot Encoding?

Representing categories as binary columns.

11. Why is data cleaning important?

To improve the quality and reliability of data analysis.

12. Which library provides MinMaxScaler?

Scikit-Learn.

Result

Thus, missing values, duplicate records, and outliers were identified and handled successfully. Data transformation techniques such as normalization and encoding were performed, and the dataset was prepared for further exploratory data analysis and machine learning applications.